

Calculus I Lesson 17

1) $f(x) = \sin x + x$

Find equation of the tangent line passing through the point (π, π) .

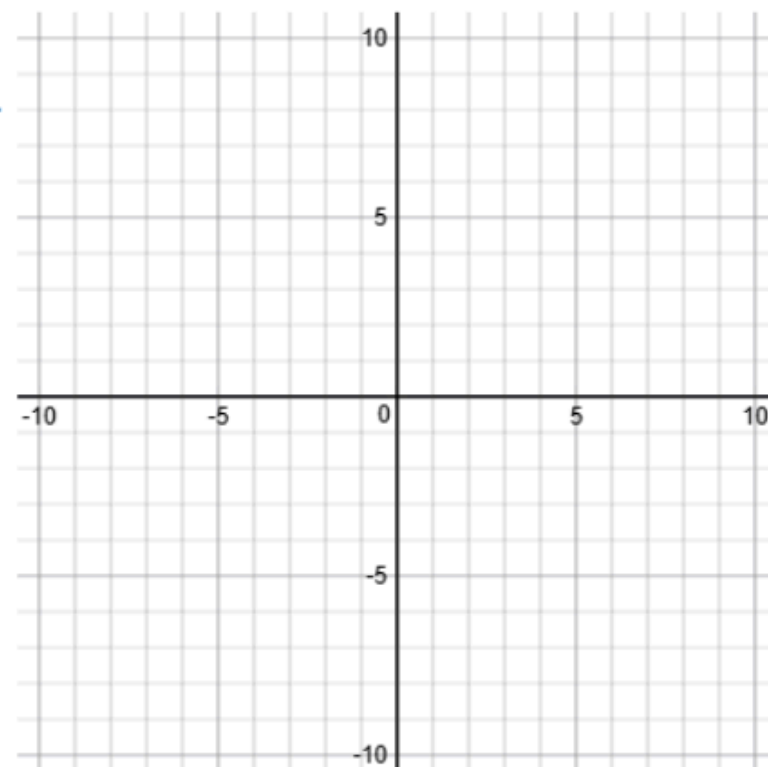
a) $f'(x) = \underline{\hspace{2cm}}?$

b) $f'(\pi) = \underline{\hspace{2cm}}?$ This is the slope of the tangent line.

c) Equation of tangent line is: $\underline{\hspace{2cm}}?$

Hint: Use $y - y_1 = m(x - x_1)$

d) Graph $f(x) = \sin x + x$ and
the tangent line at the point (π, π) .



2) $f(x) = 4 \sin x - 2x$

Find equation of the tangent line passing through the point $(\pi/2, 4 - \pi)$.

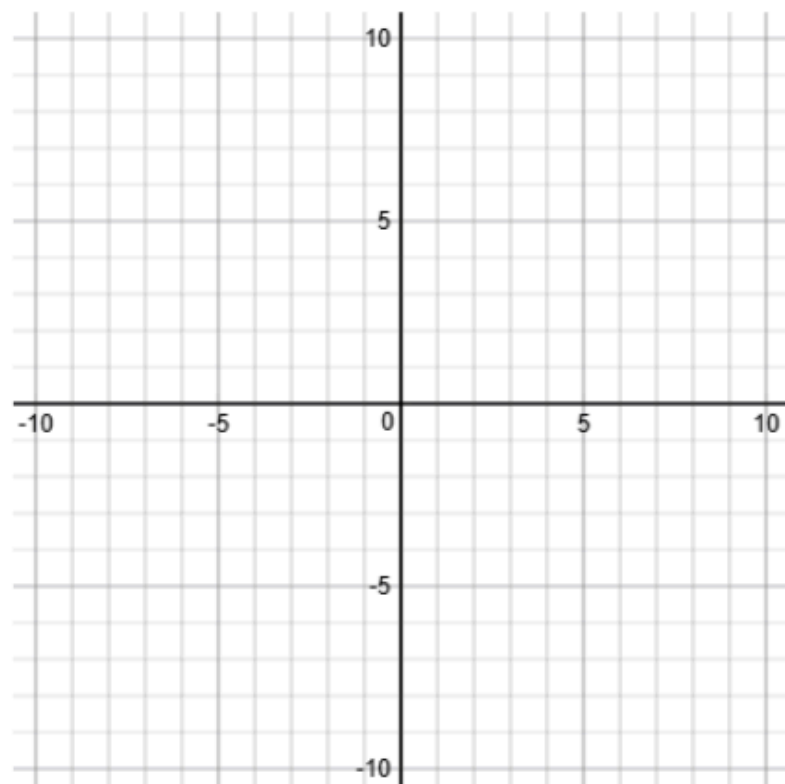
a) $f'(x) = \underline{\hspace{2cm}}?$

b) $f'(\pi/2) = \underline{\hspace{2cm}}?$ This is the slope of the tangent line.

c) Equation of tangent line is: $\underline{\hspace{2cm}}?$

Hint: Use $y - y_1 = m(x - x_1)$

d) Graph $f(x) = 4 \sin x - 2x$ and
the tangent line at the point $(\pi/2, 4 - \pi)$.



3) $f(x) = 4\cos x - 2x^2$

Find equation of the tangent line passing through the point $(0, 4)$.

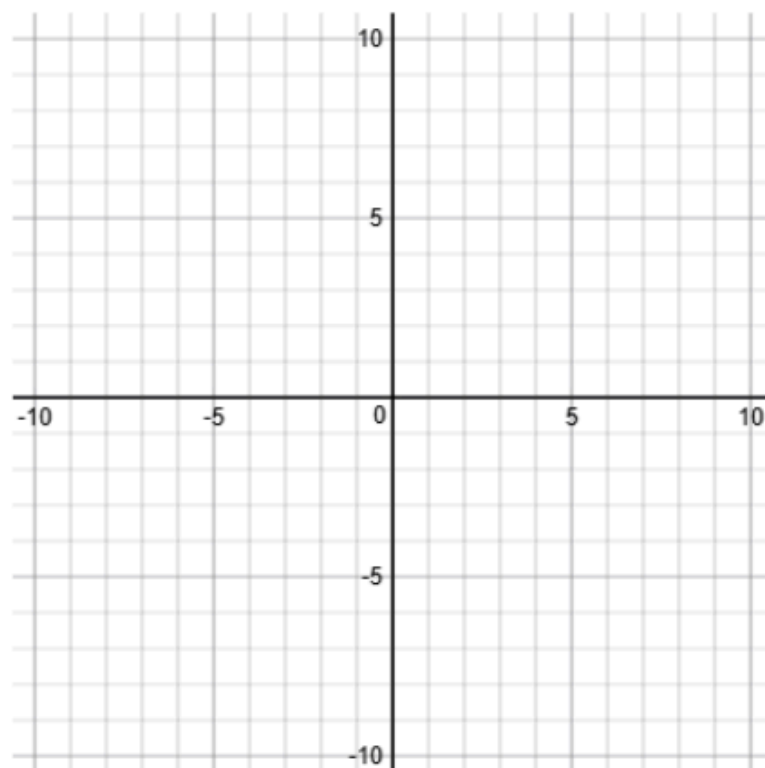
a) $f'(x) = \underline{\hspace{1cm}}?$

b) $f'(0) = \underline{\hspace{1cm}}?$ This is the slope of the tangent line.

c) Equation of tangent line is: $\underline{\hspace{2cm}}?$

Hint: Use $y - y_1 = m(x - x_1)$

d) Graph $f(x) = 4\cos x - 2x^2$ and
the tangent line at the point $(0, 4)$.



4) $f(x) = \cos x - 5x$

Find equation of the tangent line passing through the point $(\pi/2, -5\pi/2)$.

a) $f'(x) = \underline{\hspace{2cm}}?$

b) $f'(\pi/2) = \underline{\hspace{2cm}}?$ This is the slope of the tangent line.

c) Equation of tangent line is: $\underline{\hspace{2cm}}?$

Hint: Use $y - y_1 = m(x - x_1)$

d) Graph $f(x) = \cos x - 5x$ and
the tangent line at the point $(\pi/2, -5\pi/2)$.

